FIN Current-State Compendium

From the Informational Nadsoliton Postulate
to the Current Mathematical and Operational Frontier

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Purpose. This document is a self-contained, claim-graded account of the FIN research programme from its originating informational ontology through the legacy/strict kernel genealogy to the latest certified local frontier (ST461 and the current FAR state map). It presents assumptions, construction, the strongest positive and negative results, the laboratory executable specification, and the shortest honest route toward physical predictivity. It is a compendium, not a claim of physical confirmation.

Epistemic boundary. The project has no admitted independent laboratory record and no external scientific audit of the present release. All accepted results reported here are analytic, finite-dimensional, formal, interval-certified, or reproducible numerical statements within their declared models. Ontological and physical interpretations are labelled separately.

Repository. https://github.com/hyconiek/Fractal-Nadsoliton-Theory

Abstract

FIN began from a radical ontological proposal: reality has one foundation, self-similar fractal information, existing as a persistent self-coupled "nadsoliton." The founding picture further proposed that positive feedback sustains the informational excitation, that adaptive self-interaction resembles a self-learning neural network, and that light, matter and observers are emergent internal patterns. This document preserves that programme while separating it from what has actually been established. It also incorporates the later hypothesis that an observer occupies an effective refinement layer and sees deeper scales only through lossy contextual maps.

The strongest present mathematical core is a finite, dimensionless spectral-information framework. On the strict twelve-state carrier, positive radial weights define a connected weighted graph Laplacian $A = sI - W \succeq 0$. The same spectral resolution generates the unitary group $U_t = e^{-itA}$, the reversible Markov semigroup $P_t = e^{-tA}$, Green resolvents, source actions, exact Schur reductions, multiscale information geometry and conditional operational tests. Exact Green–Schur elimination rigorously realizes one form of fractal information compression, but neither produces the strict kernel from the legacy kernel nor creates an absolute length, time, action, mass or energy scale. Shannon information supports exact dimensionless identities and data-processing statements; it does not determine thermodynamic energy without an independently supplied Hamiltonian and calibration.

The historical legacy kernel is best retained as an intermediate bridge object. The later strict kernel is an operationally frozen enriched profile. A subsequent audit reconstructed the corrected Legacy* class, but did not create a third co-equal final kernel or prove a legacy-to-strict completion. The $\mathbb{Z}_{12}$ carrier, phase coherence, signed representation resources, exact process-tester optimization and a laboratory transfer specification are substantive achievements. By contrast, the historical Planck-layer hierarchy, particle-mass formulas, Standard Model identifications, gravity scaling and "maximum resonance instead of minimum energy" remain hypotheses or have been downgraded.

The latest studies strengthen both sides of this verdict. They certify twelve symmetry-related global minima for one explicitly supplied active-gain model, a first global-transition bracket $2.8934 < g_* < 2.9024964917477667$, exact low-degree invariant relations, finite observer-blind subspaces, relative clocks and multiscale obstructions. They also prove that passive spectral feedback cannot source the required gain, invariant data cannot select an orientation, and dimensionless refinements cannot choose metres, seconds or joules. The deepest interpretation surviving current falsification is therefore not a completed physical theory. It is a dimensionless spectral-information and contextual-operator research programme, with optional sourced adaptive extensions and a precisely identified missing bridge: a non-circular source law, dimensional calibration, operational realization and, when orientation matters, a selector/reference resource. A laboratory experiment has been designed as an executable specification, but no laboratory realization is claimed.

Claim vocabulary. **[PROVEN]** means a proof or exact certificate under displayed assumptions; **[STRONG EVIDENCE]** means reproducible evidence robust under the stated tests; **[MODERATE EVIDENCE]** means a plausible but incomplete inference; **[SPECULATIVE]** means a research hypothesis; **[CONDITIONAL]** means valid only after additional typed input; **[REFUTED]** means contradicted in the scope stated. These labels apply to the local FIN record, not to nature beyond the model.

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1 Executive synthesis

Present verdict. FIN has produced a coherent and unusually well-audited finite mathematical framework, but it has not yet produced a physical theory. The unifying object is a self-adjoint finite generator together with its spectral measure and contextual reductions. Information, dual dynamics, Green response and coarse graining are different functional shadows of that object. A selector, an absolute unit system, an independently sourced adaptive law and an operational realization are not shadows of the spectrum and therefore cannot be recovered from it without further axioms.

The entire programme can be read in four layers:

1. **Founding ontology.** The nadsoliton is primordial fractal information in a persistent solitonic state. There is no lower informational medium. The preferred conceptual order is

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer.}$$

   This is the theory's starting premise, not an experimentally established theorem.

2. **Finite mathematical core.** A radial kernel on a cyclic twelve-state carrier produces a real symmetric matrix, a Laplacian generator, a spectral measure, Green response, wave and heat calculi, variational source response and exact contextual compression. This layer contains the strongest proved results.

3. **Additional laws.** Learning, self-selection, orientation, physical units, preparation and measurement are not determined by the frozen kernel. They require an adaptive vector field, reference/sector data, conversion standards and an operational tuple.

4. **Physical interpretation.** Particle, field, mass, Planck-scale, thermodynamic and cosmological meanings remain conditional until a calibrated model produces held-out records. The laboratory package specifies such a test but has not supplied the record.

Table 1: The shortest current status map.

Question	Current answer	Status
What is fundamental in FIN?	One informational nadsoliton is the founding ontology; no substrate lies underneath it.	[SPECULATIVE]
What is the smallest proved core?	A finite self-adjoint generator A, its spectral measure, resolvent and declared state/context maps.	[PROVEN]
Are strict and legacy identical?	No. Legacy is an intermediate bridge profile; strict is a later enriched operational freeze. No complete map is proved.	[PROVEN]
Does one operator generate two dynamics?	Yes: e^{-itA} and e^{-tA}, with sharply different operational laws.	[PROVEN]
Is fractal compression real mathematically?	Exact Schur elimination preserves retained Green response and generates relative scale and memory.	[PROVEN]
Does it create metres, seconds or joules?	No invariant dimensionless datum can choose a point in the unit-rescaling torsor.	[PROVEN]
Does Shannon information equal physical entropy?	Only after a state, Hamiltonian, temperature/process and conversion scale are supplied.	[PROVEN]
Are particles and masses derived?	No current strict derivation supplies particle sectors, gauge representations, spin, mass scale or validated spectra.	[REFUTED] (as a present derivation)
Is a laboratory test available?	A transfer package and executable validation pipeline exist; no independent laboratory bundle has passed it.	[CONDITIONAL]

2 The founding idea and its mathematically honest reformulation

2.1 One foundation: fractal self-similar information

The originating FIN intuition is that the whole universe can be represented by one self-similar informational object. "Fractal compression" is meant to express the possibility that global structure is recurrently encoded in local structure rather than stored as an unstructured list of all events. The strongest defensible mathematical reading is not that a finite string literally contains every microscopic fact. It is that a family of effective descriptions may be generated by repeated, composable reductions of one structured operator.

The programme has now constructed one exact version of this idea. If a positive operator is partitioned into retained and hidden coordinates,

$$A = \begin{pmatrix} A_{EE} & A_{EH} \\ A_{HE} & A_{HH} \end{pmatrix},$$

then elimination of H produces

$$\mathcal{C}_H(A) = A_{EE} - A_{EH}A_{HH}^{-1}A_{HE}.$$

The retained Green response is exactly preserved. Repeating this operation yields a hierarchy of effective operators. Thus "compression" has a rigorous meaning as marginalization of response and accessible information. It does not imply lossless encoding of every possible universe in finitely many conventional bits, and it does not by itself prove physical self-similarity.

2.2 The nadsoliton as persistent information

The word *nadsoliton* names the proposed persistent, self-supporting informational state. The founding document describes a permanently excited resonance maintained by positive feedback. This idea contains three distinct claims that must be separated:

- a **persistent state**: a fixed point, periodic orbit, invariant measure or spectrally stable coherent structure;
- a **soliton**: normally a localized nonlinear travelling/coherent solution with a stability or scattering property;
- a **physical ontology**: the claim that this object is the only foundation of reality.

The current kernel theory supplies persistent spectral modes and stable finite semigroups. It does not yet supply a nonlinear continuum equation with a proved localized soliton. Consequently, persistence is mathematically available, while the full soliton and ontological claims remain **[SPECULATIVE]**.

2.3 Self-coupling, amplification and the neural-network analogy

The kernel gives a precise static form of self-coupling: each informational coordinate influences others through W_{xy}. Repeated application, resolvent summation and feedback laws describe propagated influence. Positive entries can amplify aligned components before normalization or saturation. But a frozen matrix is not a learning network by itself.

An adaptive FIN law has been studied in the general form

$$\dot{K} = -\Pi(V'(K) - C(K)),$$

where Π is a constraint projection and C is a covariance/record object. For fixed C,

$$\mathcal{F}_C(K) = \operatorname{Tr} V(K) - \operatorname{Tr}(KC), \qquad \frac{d\mathcal{F}_C}{dt} = -\|\Pi\nabla\mathcal{F}_C\|_F^2 \leq 0,$$

is a valid Lyapunov model. If $C = C(K)$, the derivative contains the additional term $DC(K)^*[K]$. Omitting it generally destroys the claimed gradient structure. The executed audit found a normalized integrability defect about 0.9048 for one tested feedback construction. Therefore:

- static network/self-coupling: **[PROVEN]**;
- a useful conditional adaptive-operator model: **[STRONG EVIDENCE]**;
- a uniquely FIN-sourced learning law or biological neural network: **[SPECULATIVE]**;
- learning derived from the strict kernel alone: **[REFUTED]**.

Neural operators and reservoir computing are good analogies because they learn maps between structured states and can exhibit memory. They do not prove that the nadsoliton learns, because training data, the loss and the update rule would replace precisely the missing physical law.

2.4 Maximum resonance versus minimum energy

The founding intuition says that the nadsoliton tends toward maximum resonance rather than a lowest-energy state. No theorem presently establishes this as a FIN law. The finite mathematics instead exposes a useful distinction.

For a symmetric coupling (W), maximizing the Rayleigh quotient

$$\mathcal{R}_W(\psi) = \frac{\langle\psi, W\psi\rangle}{\langle\psi, \psi\rangle}$$

selects a top eigenmode under a normalization constraint. For $A = sI - W \succeq 0$, minimizing $\langle\psi, A\psi\rangle$ selects the same mode ordering in the simple fixed-norm relation $A = sI - W$. Thus "maximum coupling resonance" and "minimum Dirichlet cost" need not be opposites. Their physical meanings differ only after one defines the conserved quantity, saturation, forcing and admissible state manifold.

Moreover, heat dynamics e^{-tA} damps high-A modes toward the constant zero mode, while unitary dynamics e^{-itA} preserves every modal norm. A persistent excited state requires forcing, nonlinearity, topology or a constraint not contained in the linear heat law. The maximum-resonance proposal is therefore a high-value research hypothesis, not a present conclusion.

3 Kernel genealogy: from coupling diagrams to the strict operator

3.1 The four founding coupling mechanisms

The historical diagram organized self-interaction into four conceptual pieces:

Symbolic role	Intended effect	Current interpretation
K_{geo}	geometric attenuation / viscosity	founding mechanism; its old exponential factor must not be multiplied again after the path-sum transformation
K_{res}	phase synchronization / resonance	supports an oscillatory carrier; physical resonance law remains unsourced
K_{tors}	oscillatory or torsional current	historically supplies $\cos(\pi d/4 + \pi/6)$
K_{topo}	summed paths / topology	corrected to a hyperbolic envelope $1/(1+\beta d)$ under the declared reconstruction

These are a design vocabulary, not four experimentally identified forces. Three corrections are decisive. First, multiplying an old $e^{-2.9d}$ attenuation by a second path-summed hyperbolic attenuation double-counts damping. Second, $d^{1.6}d^{-0.6} = d$, not d^{-1}; a single-path tail $d^{-2.6}$ would be needed for the stated path-counting argument. Third,

$$\cos(\pi d/4 + \pi/6) = 0 \quad \Longleftrightarrow \quad d = \frac{4}{3} + 4n,$$

so the previously quoted integer nodes (2,5,8,11) are false. These corrections do not erase the coupling intuition; they determine which algebraic form is still admissible.

3.2 Canonical legacy, strict and reconstructed Legacy*

The three names used across the historical record are best normalized as follows.

Table 2: Kernel roles after the completed audit.

Object	Formula	Scientific role
Canonical legacy	$K_{\text{legacy,ont}}(d) = \alpha_{\text{geo}} \dfrac{\cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d}$, with $\alpha_{\text{geo}} = 4\ln 2$, $\omega = \pi/4$, $\phi = \pi/6$, $\beta_{\text{tors}} = 0.01$.	Restored intermediate bridge kernel and historical ontological parameter layer. It is not a co-equal final substitute for strict.

Object	Formula	Scientific role
Strict gate	$K_{\text{strict}}(d) = \dfrac{\cos(0.18575d + 0.16250)}{1 + d^{1.8}}.$	Later gate-selected, completed/enriched finite working profile. Its phase decimals have procedural scan/refreeze provenance, not an internal physical source theorem.
Reconstructed Legacy*	$K^*_{\text{legacy}}(d) = A\dfrac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}.$	Corrected class obtained by re-auditing the four-mechanism diagram. A and β are not uniquely derived; common choices are freezes or comparison gauges. This is a corrected legacy class, not a newly proved final kernel.
Rejected product	$e^{-2.9d}\dfrac{1 + 0.2d}{1 + d}\cos(\pi d/4 + \pi/6).$	Algebraically and numerically rejected as a faithful historical reconstruction; it is already suppressed below 10^{-3} for all tested $d \geq 2$.

The strict profile is not silently derivable from the legacy formula. The missing completion data include amplitude normalization, a phase/frequency source, nonlinear compression with exponent $\eta = 1.8$, and a selector/sector source. An amplitude-only Legacy*–strict fit has relative residual at least 0.605 in the later phase audit. A canonical profile comparison gives error 0.99148; exact binary Schur reductions improve it only to 0.95117 and 0.94673. These are strong finite falsifications of a simple static compression bridge.

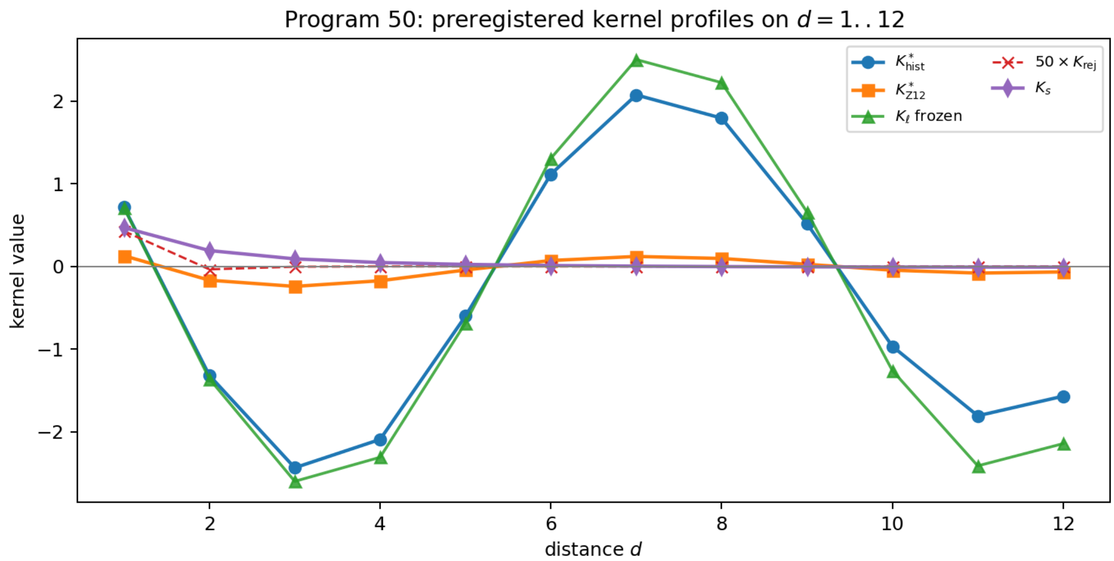

Figure 1: Audited finite profiles. Visual proximity is not a derivation. The accepted scientific question is which typed map, if any, transports legacy data into strict data without importing the target.

3.3 What "twelve octaves" currently means

The rigorous carrier is $V = \mathbb{Z}/12\mathbb{Z}$, a circle of twelve labelled states with cyclic distance. Calling these states "octaves" is an interpretation. The old argument from the three-dimensional kissing number does not currently derive a physical energy-octave system.

Legacy* has exact carrier phase period eight, whereas the finite carrier has period twelve. Their combined repeat length is $\operatorname{lcm}(8, 12) = 24$. The legacy phase is not a character of $\mathbb{Z}_{12}$. This mismatch is not an inconsistency: it defines a sampled oscillatory profile, but it blocks a naive claim of exact twelve-octave phase closure. The strict phase is compatible with the finite circulant construction, and its spectral modes are the twelve Fourier characters of $\mathbb{Z}_{12}$.

The orientation pair $+\omega$ and $-\omega$ is isospectral. Chiral receivers and bispectral markers can detect a supplied orientation, but the radial kernel does not choose its sign. This is the live selector obstruction QW-2191.

3.4 The status of the thirty fractal layers and Planck scale

Historical documents proposed thirty layers, factors β^N, and a path from Planck to macroscopic scales. These claims were based on fitted hierarchies and weakly sourced physical identifications. Current work proves something narrower:

- repeated Schur elimination generates exact *relative* scale ratios such as 2^k; **[PROVEN]**
- no dimensionless invariant can select an absolute length ℓ_*, time τ_* or action $\hbar_*$; **[PROVEN]**
- "thirty layers" is not derived by the strict operator or compression semigroup; **[REFUTED]** as a present theorem
- the Planck endpoint is an external dimensional anchor, not a strict internal unit; **[PROVEN]** as a provenance audit.

Accordingly, the compendium retains fractal layering as a future refinement hypothesis, but does not present thirty physical layers or the Planck scale as an achievement of FIN.

4 The minimal finite mathematical core

4.1 Strict weights, generator and spectrum

Let $d(x, y) = \min(|x - y|, 12 - |x - y|)$ on $\mathbb{Z}_{12}$ and set $W_{xx} = 0$, $W_{xy} = K_{\text{strict}}(d(x, y))$. The six positive off-diagonal weights are

$$\begin{aligned} &k_1 = 0.46998567264502017, \quad k_2 = 0.19204355169010282, \quad k_3 = 0.09142861427792497, \\ &k_4 = 0.04702916874565040, \quad k_5 = 0.02413122336363006, \quad k_6 = 0.011070817321442113. \end{aligned}$$

Every row has sum

$$s = 2\sum_{d=1}^{5} k_d + k_6 = 1.660307278766098953\ldots, \qquad A = sI - W.$$

Theorem 4.1 (Strict Dirichlet identity)**.** *For all* $f \in \mathbb{C}^{12}$*,*

$$\langle f, Af\rangle = \frac{1}{2}\sum_{x,y} W_{xy}|f_x - f_y|^2 \geq 0.$$

Since all off-diagonal weights are positive, $\ker A = \operatorname{span}\{\mathbf{1}\}$.

The Fourier modes diagonalize (A). Its audited spectrum is

$$0, \quad 0.754121154207080\ (\times 2), \quad 1.577049514427610\ (\times 2), \quad 1.961406861976446\ (\times 2),$$

$$2.199568849333210\ (\times 2), \quad 2.298606272079097\ (\times 2), \quad 2.342182041146301.$$

Thus the strict core has an exact finite graph-Laplacian interpretation. Calling (W) a physical interaction Hamiltonian, or (A) an energy, is an additional interpretation.

4.2 One spectrum, two dynamics

The same (A) gives

$$\boxed{U_t = e^{-itA}}, \qquad \boxed{P_t = e^{-tA}}.$$

Central dual-dynamics result. The coherent group and the diffusive semigroup are two exact Borel-functional realizations of the same spectral object (A, E_A). Their common spectrum is theorem-level. Which realization describes an experiment, and whether an open-system interpolation is present, belongs to the operational package and cannot be read from A alone.

The first is a unitary group on amplitudes. The second is a positivity-preserving, uniform-reversible Markov semigroup on probability vectors. They share projectors and eigenvalues but not operational composition of observed probabilities.

Table 3: The duality is spectral, not an identity of physical meanings.

Property	$U_t = e^{-itA}$	$P_t = e^{-tA}$
State object	complex amplitude / density operator after extra axioms	classical probability vector
Norm/invariant	Hilbert norm	total probability
Short-time escape	quadratic in (t) for position probabilities	linear in (t) when jumps are available
Interference	yes, after coherent preparation and amplitude measurement	no amplitude interference
Long-time behavior	quasiperiodic in finite dimension	converges to uniform equilibrium
Observed transition law	Born matrix generally not a semigroup	Chapman–Kolmogorov semigroup
Physical clock	not supplied by (A)	not supplied by (A)

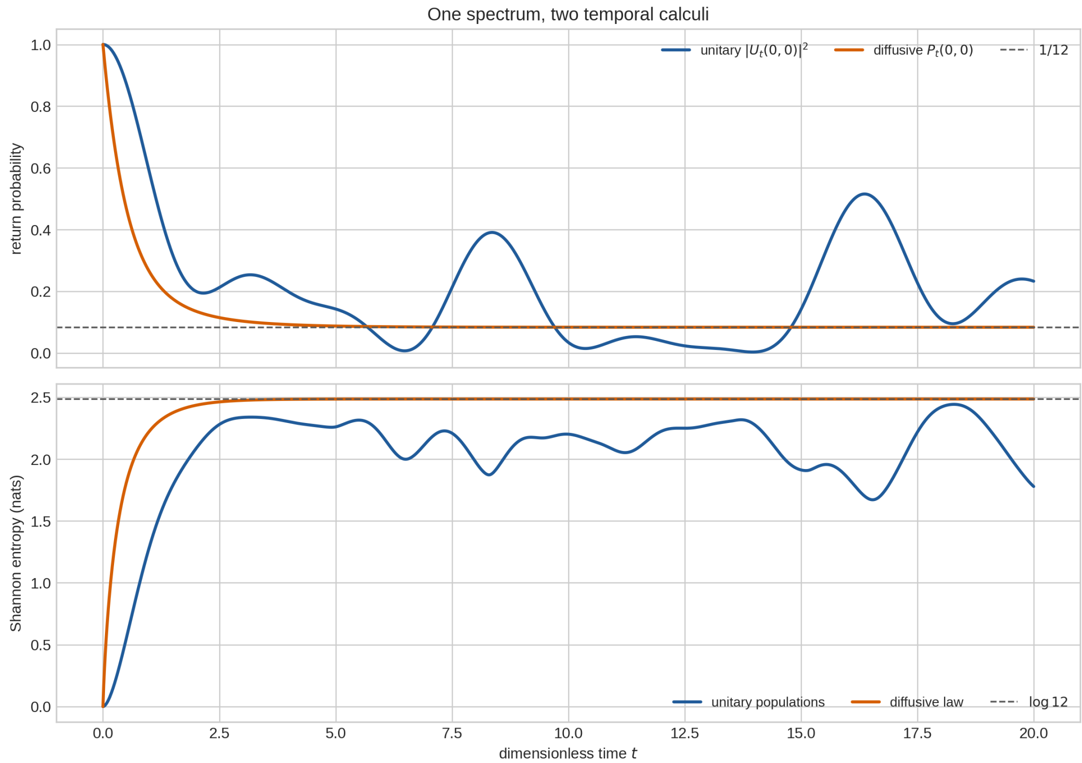

Figure 2: The strict spectrum supports coherent and diffusive temporal calculi. The choice between them, or an open-system interpolation, requires an operational model.

This duality is one of FIN's clearest achievements. It also explains the observer paradox: an internal observer who has access only to populations may mistake dephased or coarse-grained unitary motion for diffusion. Distinguishing the models requires phase-sensitive preparations, multitime interventions and recorded outcomes.

4.3 Green response and the common spectral object

For $z \notin \sigma(A)$,

$$G(z) = (zI - A)^{-1} = \int \frac{1}{z-\lambda}\, dE_A(\lambda).$$

The same projector-valued measure E_A generates $G(z)$, U_t, P_t, spectral filters and source response. This is the precise sense in which many repository structures are shadows of one object.

The Green resolvent also supports a corrected field-theoretic reconstruction. Given a supplied positive $L + m^2 I$,

$$S[\phi, K] = \frac{1}{2}\phi^T (L + m^2 I)\phi - J^T \phi + \operatorname{Tr} V(K) - \operatorname{Tr}(KC)$$

has stationary equation

$$(L + m^2 I)\phi = J, \qquad \phi = (L + m^2 I)^{-1} J.$$

This reconstructs a minimal quadratic action from a propagator. If (V'(K)=C), it also gives a conditional operator potential. However, fitting (V) after observing a target (K) is always possible by finite polynomial interpolation and has no predictive force. The action is therefore a valid inverse construction, not yet a uniquely FIN-sourced fundamental Lagrangian.

Some earlier referee language classified strict as a normal-ordered screened discrete Green/Yukawa-type object. The robust surviving statement is the finite resolvent/Green interpretation. A unique continuum Yukawa field, metric, source law and renormalization prescription have not been derived.

5 Fractal compression, memory and the observer

5.1 Exact Green–Schur compression

Partitioning visible (E) and hidden (H) states gives the frequency-dependent effective operator

$$A_{\text{eff}}(z) = A_{EE} - A_{EH}(zI + A_{HH})^{-1}A_{HE},$$

with self-energy

$$\Sigma_E(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}.$$

At zero frequency this reduces to the Schur complement. Eliminating alternating vertices in the tested $24 \to 12$ construction preserved retained Green response to residual 1.44×10^{-14}. This is an exact finite compression theorem, not a visual analogy.

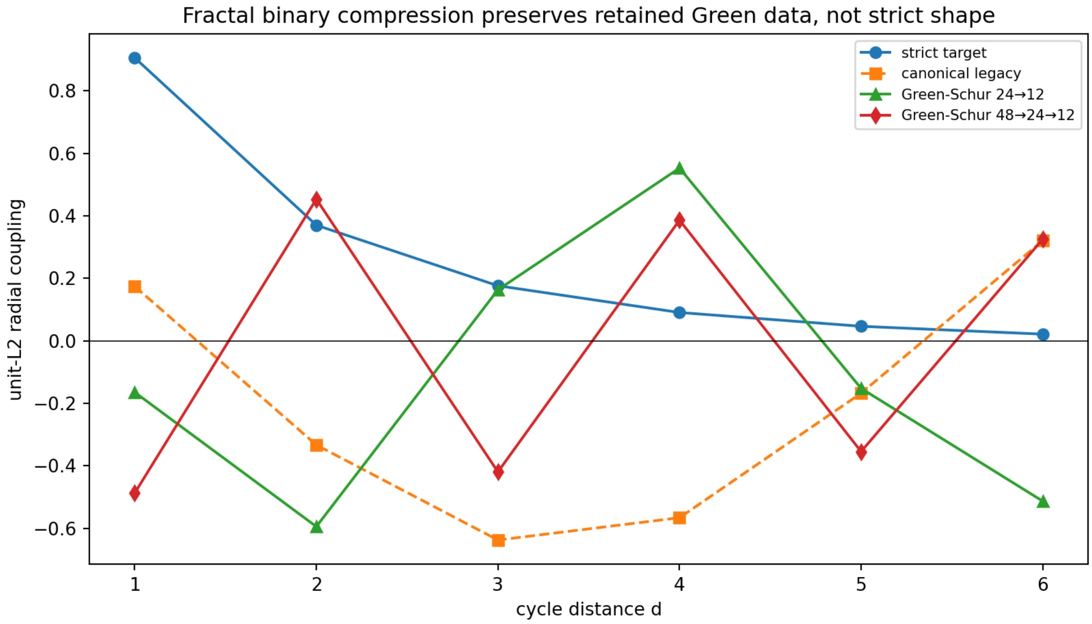

Figure 3: Exact response-preserving elimination generates a hierarchy of effective descriptions. It gives relative scale and hidden-state memory, but did not generate the strict kernel in the executed tests.

The dynamic self-energy is a Stieltjes-type matrix function. Its time transform is a memory kernel, connecting Green functions, Mori–Zwanzig/Feshbach reduction, adaptive reservoirs and contextual observation. The second-generation atlas identified the *Stieltjes–Schur Context Functor* as the most informative derived object:

$$\text{Green response} \leftrightarrow \text{compression} \leftrightarrow \text{memory} \leftrightarrow \text{observer context} \leftrightarrow \text{accessible information.}$$

5.2 What compression explains and what it does not

Compression explains how hidden degrees of freedom modify visible laws, create nonlocal memory and reduce distinguishability. In a regularized Gaussian information model, the legacy-versus-strict divergence fell from (23.63624) to (11.60157), consistent with data processing. This is apparent information loss from the retained context, not ontological deletion.

Compression did *not* produce $\eta = 1.8$, the strict phase, a physical length or the missing orientation sign. Its exact achievements are therefore substantial but bounded:

- exact marginal response and composable coarse graining: **[PROVEN]**;
- dynamic memory/self-energy space: **[PROVEN]**;

- a nontrivial fractal fixed point in that function space: **[SPECULATIVE]**;
- strict as a static fixed point of binary compression: **[REFUTED]** in the tested family;
- absolute physical scale from fractal ratios: **[REFUTED]**.

5.3 Information loss, negative coupling and signed resources

The phrase "negative information coupling" admits at least three inequivalent meanings:

1. a negative scalar kernel entry;
2. contraction of accessible distinguishability after a channel or context restriction;
3. signed mass needed to represent a target attenuation as a mixture of simpler paths.

Only the second is automatically an information-theoretic loss. The executed programs proved that a larger reversible description can retain information that disappears from the observed subsystem. The raw signed legacy operator is not a Markov generator; absolute-value, positive-part or conservative-shift repairs must be declared.

For the strict attenuation sequence $h_k = (1 + k^{9/5})^{-1}$, every positive exponential-distance mixture obeys Hausdorff moment inequalities, but $\Delta^4 h_0 = -0.0715275581$. A signed representation is therefore necessary. The exact seven-moment minimum negative Jordan mass is

$$\mathcal{N}_{\text{path}}^{\min} = 0.406706334692258\ldots,$$

with a certified interval lower endpoint (0.4067063265581317). For the full oscillatory twelve-moment profile the later exact enclosure was narrowed to approximately

$$0.7073534379053974 \le \mathcal{N}_{\text{osc}} \le 0.7073534683998260.$$

These are rigorous classical signed-representation costs. They are not negative energy, quantum negativity, information destruction or laboratory loss.

6 Phase coherence, symmetry and the missing selector

6.1 What phase results are established

The project contains several genuine phase results:

- the legacy phase $\pi d/4 + \pi/6$ is exact and has period eight in its carrier;
- the strict frozen decimals are generated by the procedural lattice $1/4000$, with $0.18575 = 743/4000$ and $0.16250 = 13/80$;
- finite Fourier projectors give exact modal coherence on $\mathbb{Z}_{12}$;
- chiral bispectra and inversion-odd receivers distinguish a supplied (+/-) sector;
- Schur compression transports the chiral receiver while preserving the mirror relation.

The $1/4000$ unit has scan/refreeze provenance. It is not an internally derived physical phase quantum. The exact identity $\alpha_{\text{geo}} = 4\ln 2 = \ln 16$ and

$$A_\phi = \frac{2\pi}{\alpha_{\text{geo}}} = \frac{\pi}{2\ln 2}, \qquad \alpha_{\text{geo}} A_\phi = 2\pi,$$

is mathematically exact. Since $\alpha_{\text{geo}}/(2\pi)$ is irrational, it does not yield a finite cyclic clock or a canonical orientation.

6.2 Why mirror symmetry is not broken internally

On a radial $\mathbb{Z}_{12}$ kernel, inversion $d \mapsto -d$ leaves the spectrum unchanged. Exhaustive audits of automorphism-invariant Boolean predicates and orbit-constant scoring functionals failed to distinguish $+1$ from -1. This is not a lack of cleverness but a no-section obstruction: an invariant object cannot canonically choose between symmetry-related branches.

An apparatus orientation, boundary condition, chiral state or perturbation can select a branch. That is a valid [**CONDITIONAL**] physical model. It is not a strict-core derivation. Consequently no current result discharges QW-2191, and no old oriented physical role may be transferred from legacy to strict without a separate role-transfer theorem.

7 Shannon information and the bridge to thermodynamics

7.1 What the four-bit identity proves

For a uniform sixteen-state source,

$$H = \ln 16 = 4\ln 2 = \alpha_{\text{geo}}.$$

This is exactly four bits in natural logarithmic units. It gives a compact informational normalization candidate and explains why $4\ln 2$ recurs in the legacy programme. It does not uniquely determine a kernel amplitude, a physical entropy or a mass.

For any stochastic channel T, relative entropy satisfies data processing,

$$D(p\|q) \geq D(Tp\|Tq).$$

This is the rigorous foundation for the FIN accessible-information ledger. Unitary position probabilities can exhibit information backflow because the population map is not a Markov semigroup; the executed strict/Legacy* tests found monotone Markov information and numerous unitary Born backflow steps.

7.2 Why information is not yet energy

Every strictly positive finite probability vector can be written as a Gibbs distribution at infinitely many energy scales:

$$E_i = -\theta \log p_i + C, \qquad \beta = \theta^{-1}.$$

Changing θ changes all energy gaps while preserving p. A modular Hamiltonian $-\log\rho$ is likewise dimensionless until time or temperature is calibrated. Landauer's equality can be reproduced after a Hamiltonian, bath, temperature and erasure process are supplied; it does not assign energy to a bit in isolation.

The minimal conditional thermodynamic bridge is therefore a tuple containing at least a state, Hamiltonian, inverse temperature, allowed processes and an energy/action calibration. FIN currently supplies the informational side of this bridge, not its physical scale.

8 Can the soliton's frequency generate time?

The intuition that a persistent oscillation carries time is scientifically useful. Mathematically, a nonzero spectral gap $\widehat{\omega}$ supplies a dimensionless phase parameter

$$\varphi(t) = \widehat{\omega}t \pmod{2\pi}.$$

If a mode is persistent and observable, its phase can define an *internal relational clock*: other processes can be compared with its cycles. Ratios $\widehat{\omega}_i/\widehat{\omega}_j$ and cycle counts are intrinsic.

Three steps remain external:

1. choose which dynamics and which mode is the clock;
2. prove persistence, readability and sufficiently small drift under the declared environment;
3. calibrate one cycle against a physical second, or supply an equivalent dimensional reference.

Rescaling $t \mapsto ct$, $A \mapsto A/c$ preserves all dimensionless phase relations. Therefore the spectrum can generate order, periodicity and relative duration but not an absolute second. The user's intuition is best promoted to the following local research conjecture:

> **Relational clock conjecture.** A persistent nadsoliton mode plus an operational record functor can define an internally observable time coordinate up to positive affine rescaling. Absolute time units remain a calibration resource.

This conjecture is compatible with the current obstruction and can be investigated without claiming that time in nature has already been derived.

9 Matter, light and old physical-role claims

9.1 The historical emergence picture

The founding programme proposed that particles are stable topological defects or self-sustaining informational domains, that mass measures the cost of changing such a pattern, and that light and matter arise as modes of one field. This is a coherent research narrative. It resembles nonlinear field defects, Hopfions, information-processing cost and emergent quasiparticles.

However, the current finite kernel does not provide the mathematical objects required to make those identifications unique: a continuum/refinement space, a nonlinear field equation, boundary conditions, a topological charge, gauge group and representations, spin/statistics, dimensional action and a calibrated observable map. A finite twelve-state radial matrix alone cannot prove that one eigenmode is an electron or that a winding number is a particle generation.

9.2 Audit of the principal historical physical claims

Table 4: Status of historical physics claims.

Claim	Current status	Reason
Particles are topological defects of the nadsoliton	**[SPECULATIVE]**	No strict continuum field, defect class or stability theorem connects the finite kernel to particles.
Mass = topology × resonance / processing intensity	**[SPECULATIVE]**	Historical correlations and fitted formulas are not a noncircular derivation; the mass unit and topological assignment are missing.
Four-bit jumps explain lepton/proton masses	**[REFUTED]** as a derivation	Approximate post-hoc fits and ad hoc assignments do not identify unique particle sectors or held-out predictions.
$\sin^2\theta_W = \alpha_{\text{geo}}/12$	heuristic only	The formula is not a proved operator functional and has no strict role-transfer certificate.
$\alpha_{\text{EM}}^{-1} = \alpha_{\text{geo}}(1-\beta_{\text{tors}})/(2\beta_{\text{tors}})$	model-level / conditional	It depends on a historical parameter coordinate and fails the required gauge/role-transfer test.

Claim	Current status	Reason
Gravity hierarchy β^N and thirty layers	**[SPECULATIVE]**/ downgraded	A dimensionless fitted hierarchy does not create Newton's constant, Planck length or a sourced layer number.
Strict kernel gives Standard Model, GR or quantum gravity	**[REFUTED]** as a present claim	Gauge, spin, Lorentzian, metric, stress–energy, continuum and unit structures are absent.
Nadsoliton $\to$ light $\to$ matter $\to$ observer	ontology/program order	It remains the preferred FIN emergence programme, not a completed derivation.

The honest answer to whether the old matter programme "had any sense" is therefore twofold. It had legitimate heuristic value: stable modes, nonlinear defects and internal observers are standard mathematical mechanisms worth testing. Its specific particle identifications and mass formulas do not survive as established results. This compendium retains the research questions and rejects their promotion to physics.

10 The missing physical structure

10.1 Why the operator is not a complete theory

A self-adjoint matrix does not specify what is prepared, how time is calibrated, what is measured, which environment is ignored, what apparatus defines orientation, or what event becomes a record. The smallest current architecture is

$$\boxed{W_0 + \mathrm{CA} + \mathrm{OA} \quad (+\mathrm{SA} \text{ when orientation matters})}.$$

Here:

- W_0: the dimensionless informational core — carrier, kernels with distinct roles, A, spectral measure, Green response, compression and information geometry;
- CA: conversion axioms, minimally a basis equivalent to $(\ell_*, \tau_*, \hbar_*)$;
- OA: operational axioms
$$(\rho_0, \mathcal{P}, \mathcal{T}, \mathcal{I}, \mathcal{E}, \mathcal{A}, \mathcal{R}),$$
for state, preparation, dynamics/clock, instrument, environment, apparatus and record;
- SA: a reference origin/orientation/chirality and its coupling law when the outcome is sign-sensitive.

The rank-three unit basis is necessary because independent rescalings of length, time and action leave the dimensionless FIN core invariant. Removing one leaves at least one of length, time, energy, mass or action underdetermined.

10.2 Operational completion by processes and testers

Process tensors/quantum combs provide the most developed mathematical home for the missing observer structure. A comb represents multitime dynamics with memory; a tester represents allowed preparations and instruments; classical outcome registers represent records. Clock and reference calibration determine how the abstract slots map to laboratory operations. This framework can distinguish coherent, Markov, dephasing and non-Markovian descriptions of the same spatial operator.

It does not create a unique apparatus from the kernel. It tells us how to state and falsify one proposed apparatus. This is exactly the boundary between a mathematical model and empirical physics.

11 The constructed experiment and executable specification

FIN Release 10.21 constructed a transfer package rather than claiming an experiment had occurred. Its five compulsory measurement obligations are:

1. basis and informationally complete preparations;
2. a calibrated time variable;
3. a vertex-resolving POVM;
4. finite event-level transition counts;
5. an independent record, frozen hold-out and separation of provider, registrar and analyst.

The first four can be specified and validated by mathematics/software. The fifth cannot be generated by code in the same custody chain.

11.1 P240–P242

- **P240** freezes a twelve-state heat-kernel tomography design. Its optimal dimensionless acquisition time is $\tau_* = 1/\lambda_{\max} \approx 0.426952$, with allocation and finite-shot guidance.
- **P241** is a fail-closed validator for preparations, clock, vertex measurement, event data, hashes and custody roles.
- **P242** is a one-shot held-out pipeline for semigroup and projective-spectrum tests. It may not run before the external gate passes and must report failure without repairing the model after unblinding.

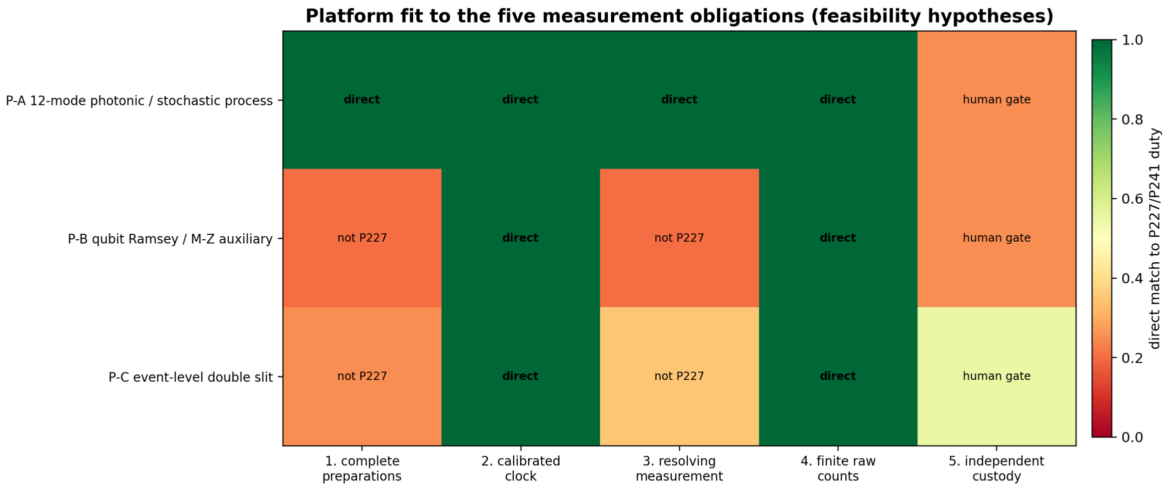

Figure 4: The operational obligation matrix. This is an executable specification of what a physical test must supply, not evidence that the apparatus exists.

11.2 Candidate platforms

The primary feasibility hypothesis is a photonic continuous-time walk on a twelve-mode graph, with single-photon or weak coherent preparation, an interferometric/waveguide implementation, controlled dephasing/loss, vertex-resolving detectors and time-tagged events. An interferometric qubit/Ramsey platform is an auxiliary coherent/dephasing test. An event-level double slit is a custody and coherence control and must store raw tuples $(t, x, y, \text{configuration}, \text{run id})$, not only an image.

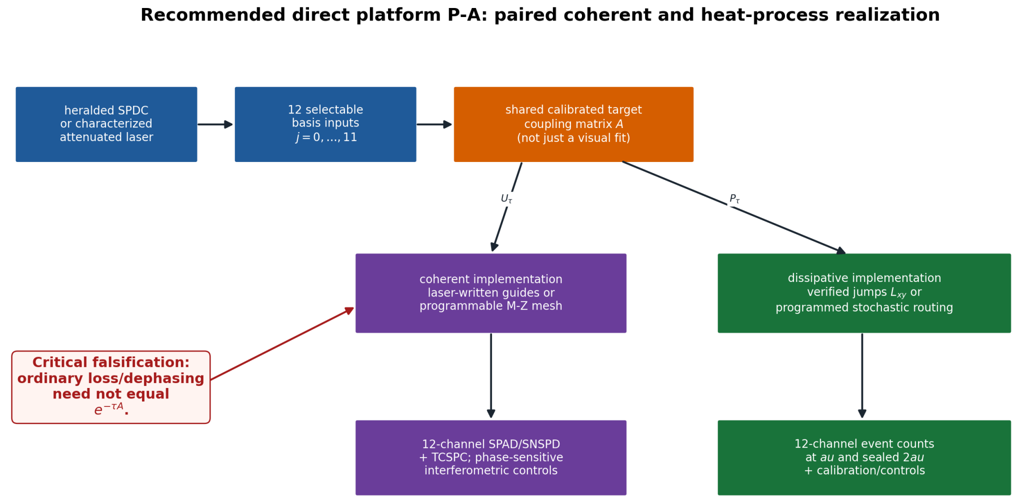

Figure 5: Candidate twelve-mode photonic realization. Every arrow is a hypothesis to be certified by experimental physicists.

The finite double-slit model predicts an interference cross term for coherent two-vertex preparation and its suppression by a which-path record. It is a valid finite operational analogue, not a simulation of continuum optical diffraction and not by itself a test of the twelve-state heat semigroup.

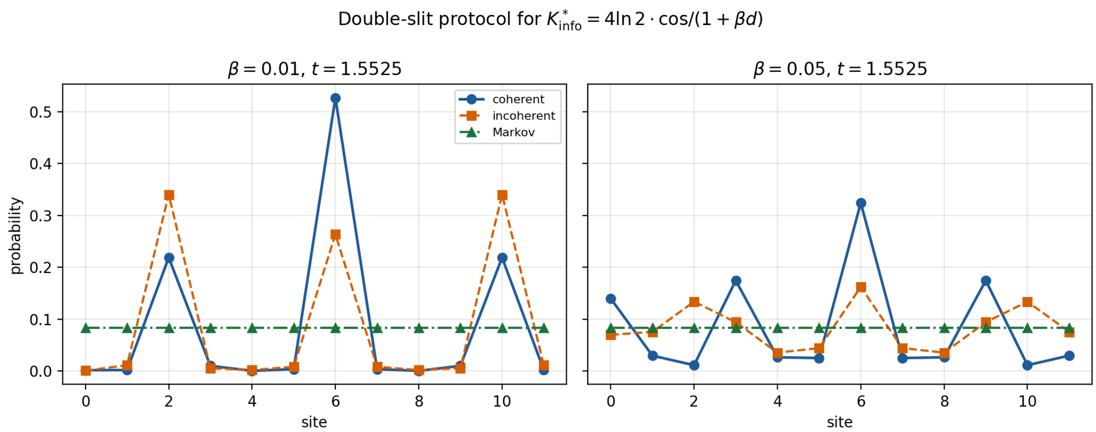

Figure 6: Finite event-model double-slit comparison. The figure illustrates a model-discrimination observable; it is not laboratory data.

No external bundle has passed P241. Hence the current statement is: [CONDITIONAL] a falsifiable experiment has been designed; **[REFUTED]** the claim that FIN is already laboratory validated.

12 Later research: signed resources and exact operational mathematics

12.1 From P295 to P393

Later local programmes sharpened the legacy–strict boundary and the operational interface:

- strict attenuation cannot be a positive fixed-phase mixture of legacy exponential paths; exact signed-resource lower bounds were certified;

- the legacy finite phase group and strict procedural nontorsion phase resource cannot be connected by a torsion-preserving map;
- scalar finite interpolants can map one C_{12} spectrum to another but fail transfer to other carriers, proving interpolation is not universality;
- a six-outcome informationally complete finite POVM, loss-aware synthetic photonic model and multitime testers were constructed;
- FIN Operational Resource Completion (FORC) typed the missing preparation, time, measurement, environment and record resources;
- exact operational realizations of signed sampling were constructed with ordinary probabilities and a sign register, without interpreting the signed measure as negative physical probability;
- no independent hold-out, calibrated hardware, physical reservoir or dimensional standard was admitted.

These results make the framework more falsifiable and reproducible. They do not complete the physical bridge.

12.2 The P394–P486 exact causal-tester frontier

The most advanced current local mathematics concerns reduced multitime phase/dephasing discrimination. It is downstream of the FIN operator and should not be mistaken for a kernel derivation.

- P451 proved a coherent tester advantage over the complete diagonal face.
- P454–P469 reduced a global primal–dual gap through exact nested dual certificates and support/KKT structure.
- P471–P473 reduced the support condition to thirteen cubic polynomial orbits, proved a strict rational Krawczyk inclusion and positivity, and certified exact global attainment of one O167 optimizer for the declared reduced three-slot problem. The value lies in

$$[0.5233281002710117,\ 0.5233281002710717].$$

- P483 proved persistence in a small parameter rectangle. Root uniqueness is local to the certified box.
- P474/P484 found strong evidence for a complex optimal-face direction but also an exact counterexample showing that Riccati congruence and orientation alone do not force the missing causal axis. Full-cone nonuniqueness remains open.
- P480 independently replayed the exact interval certificate with the Python standard library.
- P486 certified determinant signs, the positive standard branch, a nonzero pivot and $\det(B_{\mathrm{map}}) = +1$. It did not prove the causal equal-endpoint identity.

This is genuine computer-assisted mathematics. It supplies no selector, unit, laboratory record, legacy-to-strict completion, particle theory or gravity.

13 The twenty-hour Singular/Kaggle checkpoint

The supplied archive `FIN_Singular_Checkpoint_after_PREPARE.zip` was independently inspected for this compendium. Its SHA-256 is

`a55500e30a194cb16ced94266c17e29e5b555669e064fcbda242c52a3f535a67`.

It contains an exact rational input object of about (12.7) MB, a manifest, a prepare-status file and campaign state. The decisive facts are:

- status: `prepared_exact_QQ_system`;
- eleven reduced variables and eleven reduced generators;
- five P485 target polynomials;
- coefficient closure $\sqrt{2} = 2 - 4\alpha^2$ for $\alpha = \sin(\pi/8)$;
- no literal `sqrt` remains in the prepared rational system;
- the recorded preparation stage took about (1170.72) seconds;
- the largest generator has about (1.85) million characters;
- no P485 Gröbner basis, no five normal-form remainders and no P487 elimination relation are included.

The checkpoint is therefore a valuable reproducible input artifact and proves that the earlier $\sqrt{2} \to \mathbb{Q}$ implementation defect was repaired at the preparation layer. It is not a P485 or P487 scientific verdict. The user's longer Kaggle wall time may include queueing or a later computation, but the archive itself certifies only PREPARE; this report does not infer missing results.

14 The post-compendium research wave: ST278–ST461

Release 10.48 ended before a long sequence of local analytic and certified-computational studies. The present section integrates their strongest conclusions. These programmes did not change the frozen strict kernel. They asked a more disciplined question: after supplying one explicit nonlinear or operational extension, which structures can be proved, which remain underdetermined, and which are impossible within the stated class?

14.1 A supplied active-gain model and its exact limitation

The central nonlinear test family is defined on the probability simplex

$$\Delta_{11} = \left\{ p \in \mathbb{R}^{12} : p_i \geq 0, \ \sum_i p_i = 1 \right\}, \qquad u = \frac{1}{12}\mathbf{1},$$

by the dimensionless functional

$$V_g(p) = D(p\|u) - \frac{g}{2}(p-u)^T A(p-u), \qquad D(p\|u) = \sum_i p_i \log \frac{p_i}{u_i}. \tag{1}$$

The entropy term penalizes localization; the second term rewards structure detected by the strict generator. This is the cleanest current mathematical realization of "coherent informational reinforcement." It is also an *extension*: the scalar gain g and the active sign in (1) are supplied, not derived from the frozen kernel.

Two source results prevent circular interpretation.

Proposition 14.1 (Passive-source obstruction)**.** *Within the tested passive functional-calculus and passive-memory classes, nonnegative spectral feedback cannot export the negative quadratic term required in* (1)*. An active pump, signed auxiliary sector or equivalent resource is necessary.*

This establishes an important boundary. Symmetry creates a degenerate family of possible branches, but symmetry is not itself positive feedback. The dynamical chain must be written

$$\begin{aligned} &\text{symmetry} \longrightarrow \text{degeneracy} \longrightarrow \text{unstable or neutral mode} \longrightarrow \text{selection} \\ &\qquad \longrightarrow \text{feedback} \longrightarrow \text{stable phase.} \end{aligned}$$

The gain supplies the feedback step. FIN does not yet source it.

For the supplied family, however, the results are strong. The uniform state is a local minimum until the Hessian threshold

$$g_{\mathrm{Hess}} = \frac{12}{\lambda_{\max}(A)} = 5.123427551\ldots.$$

Nevertheless a nonuniform branch becomes globally favorable earlier: this is a first-order transition with metastability, not a loss of local stability of u.

Theorem 14.2 (Certified first global transition bracket)**.** *For the declared strict A and the functional (1), the first value at which the uniform state ceases to be the unique global minimizer satisfies*

$$\boxed{2.8934 < g_* < 2.9024964917477667}.$$

The lower bound follows from a rigorous entropy inequality and a 224,626-leaf adaptive cover; the upper bound is supplied by an explicit certified competitor.

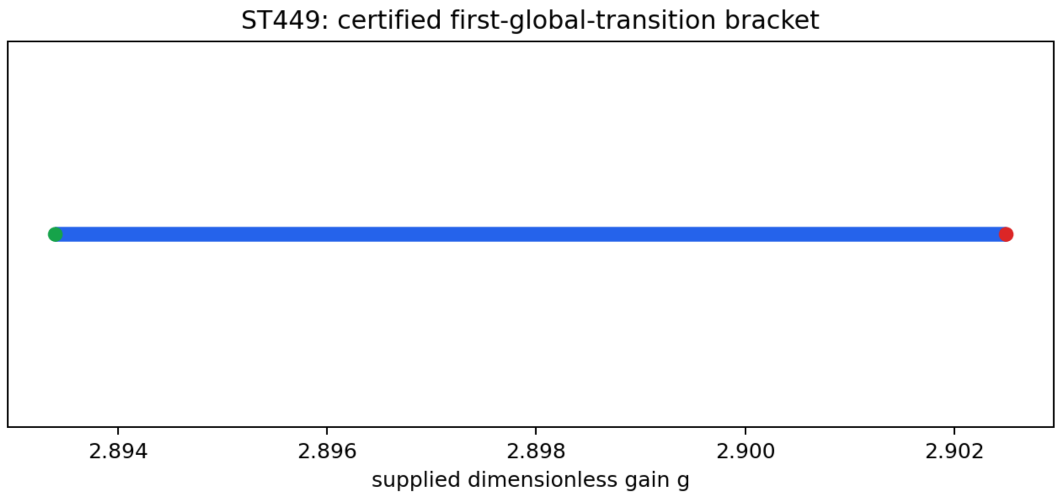

Figure 7: The certified first-transition bracket for the supplied active-gain variational family. The figure proves neither a FIN source for g nor a physical critical point.

At the particular supplied value $g = 4$, interval and symmetry arguments prove exactly twelve global minima forming one orbit under D_{12}. This is an exact example of a symmetric potential supporting twelve equivalent realized phases. It does *not* select one member of the orbit. A prepared state, fluctuation history, boundary condition or reference resource is still required to say which branch occurs.

A separate local event was isolated in

$$[2.902496471747767,\ 2.9024964917477667].$$

The full 17-dimensional Jacobian has one certified null direction there, and both fold-transversality coefficients are nonzero. This proves a local simple saddle-node for the declared reflection-even branch. It does not identify the first global minimizing orbit or prove uniqueness of the global transition.

14.2 Stationary atlas, Morse data and the infrared branch

The stationary-point campaign certified at least 85 distinct stationary points in the declared symmetry reduction. Its current partial Morse polynomial is

$$M_{\mathrm{partial}}(t) = 13 + 42t + 30t^2 = 1 + (1+t)(12+30t).$$

This factorization is compatible with the simplex topology but is not a complete Morse census. Missing higher-index or boundary points have not been excluded globally.

The infrared continuation campaign then constructed overlapping interval/Krawczyk cells and boundary links for a connected local branch across the complete tested interval $0 \le b \le 0.1$. The local degree remains -1. This is a robust existence/continuation result for the stated equations; it is not a continuum limit, a physical renormalization group or a proof that the branch is globally unique.

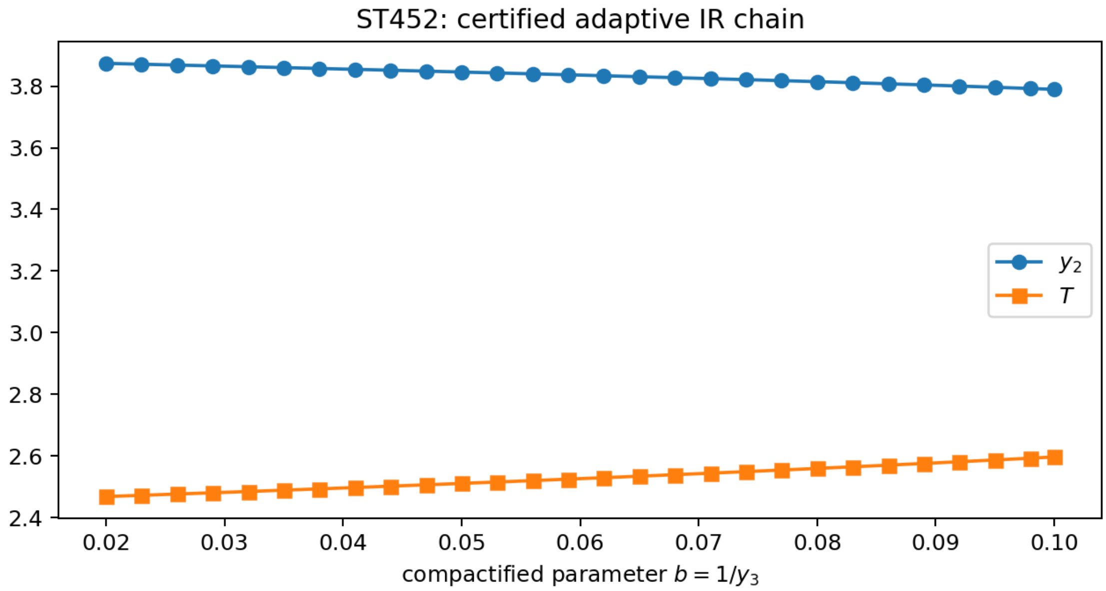

Figure 8: Certified local continuation of the infrared branch. Connected local coverage must not be promoted to a global root count.

These studies make the word "persistent" more concrete: one can certify stable or stationary branches after an explicit nonlinear law is declared. They do not yet establish a travelling, localized, scattering-stable soliton in a refinement or continuum family. Thus they support a *nadsoliton research mechanism*, not the complete nadsoliton ontology.

14.3 Invariant algebra: what symmetry permits and what it cannot source

Exact modular and integer computations substantially clarified the D_{12} invariant algebra.

- Sixteen primitive integer relations of degree six were exported in the declared invariant basis. Their exact characteristic-zero representatives are divisible by the translation-mode factor $\sum_i x_i$; the decomposable span has rank 310 and the reported quotient dimension is 55.
- A projective-incidence calculation over two large primes tested 1,337 of 1,390 projected triples and proved that no rational degree-eight relation with support at most three exists in the declared design. Since dimension counting forces at least 136 relations, those relations must involve support at least four.
- A complete equivariant module calculation identifies which polynomial terms are allowed by D_{12}. Allowed is not the same as generated: symmetry classifies a candidate law but does not supply its coefficient, sign or physical provenance.

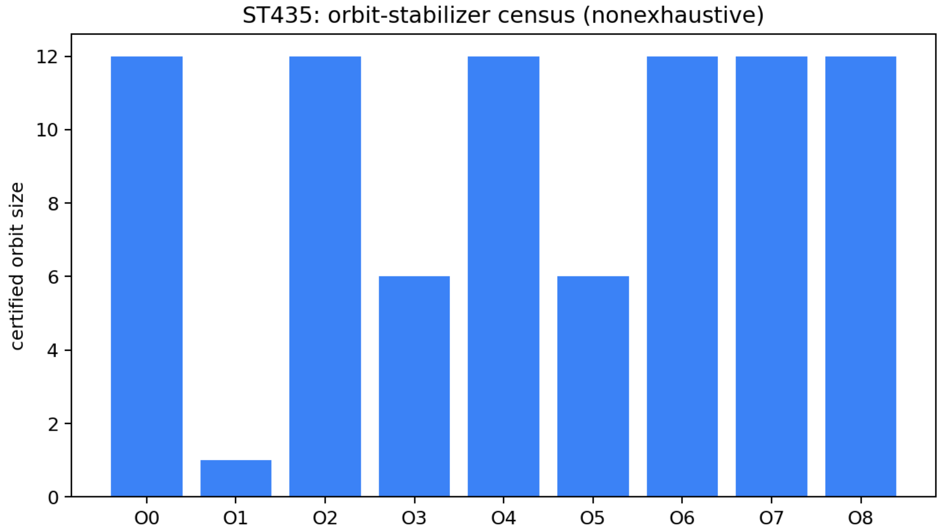

Figure 9: Symmetry-orbit structure in the stationary atlas. Orbit multiplication explains repeated solutions but does not select an orientation or physical vacuum.

One entropy expansion gives the first allowed angular anisotropy

$$V(r,\varphi) = V_0(r) + \frac{7776}{11} r^{12} \cos(12\varphi) + O(r^{14})$$

in the declared local coordinates. This proves that twelvefold anisotropy can appear at high order once the state and functional are specified. It does not derive the active gain, the phase origin or a unique orientation from the strict radial kernel.

14.4 The internal observer, refinement layers and relational scale

The later studies give a precise mathematical form to the user's intuition that an observer may live inside one developed layer of a fractal hierarchy and see deeper scales only through successive coarse maps.

Let X_n be a fine state space, $C_{n\to m}$ a declared coarse channel and $\mathfrak{I}_m$ the instruments accessible on layer m. Two fine states are operationally equivalent for that observer if

$$\rho \sim_m \sigma \quad \Longleftrightarrow \quad \mathrm{Tr}[M\, C_{n\to m}(\rho)] = \mathrm{Tr}[M\, C_{n\to m}(\sigma)] \quad \text{for every } M \in \mathfrak{I}_m.$$

This quotient is an observer-relative description, not a second informational substrate below the nadsoliton.

The executed refinement models contain a 78-dimensional coarse-blind kernel: ordinary coarse measurements cannot distinguish those fine directions. A specially enriched child-level tomography can recover the hidden block. Thus

$$\text{fine state} \longrightarrow \text{coarse record}$$

can be many-to-one even when the full theory is finite and deterministic. This rigorously supports the statement "the observer sees lower scales through layers," provided it is understood as restricted operational access rather than literal visual transparency.

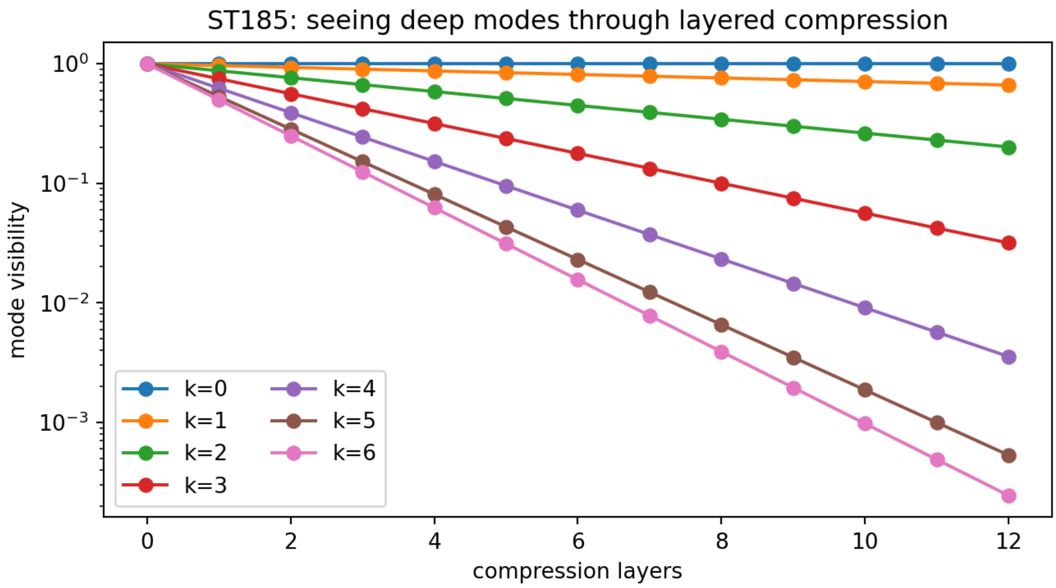

Figure 10: Layer-relative visibility: coarse instruments identify equivalence classes, while enriched instruments can separate hidden fiber degrees of freedom.

Physical units are likewise relational at the level presently derived. The conversion standards form a free calibration torsor,

$$(\ell_*, \tau_*, \hbar_*) \mapsto (a\ell_*, b\tau_*, c\hbar_*), \qquad (a, b, c) \in (\mathbb{R}_+)^3.$$

Invariant FIN data determine ratios, not a canonical section of this torsor. Adding a speed and an action scale reduces the freedom but leaves one positive orbit; a third independent anchor is required for the full action-density dimension. This is the minimal-unit result, not a defect of notation.

An internal spectral clock can therefore compare modes,

$$\frac{\omega_r^{(U)}}{\omega_s^{(U)}} = \frac{\lambda_r}{\lambda_s}, \qquad \frac{\omega_r^{(C)}}{\omega_s^{(C)}} = \sqrt{\frac{\lambda_r}{\lambda_s}},$$

but cannot create a second. The unitary scaling $(A, t) \mapsto (cA, t/c)$ preserves phase trajectories, whereas the wave scaling is $(A, t) \mapsto (cA, t/\sqrt{c})$. Channel-specific maps must therefore be declared before one spectrum is assigned physical frequency, relaxation and wave units.

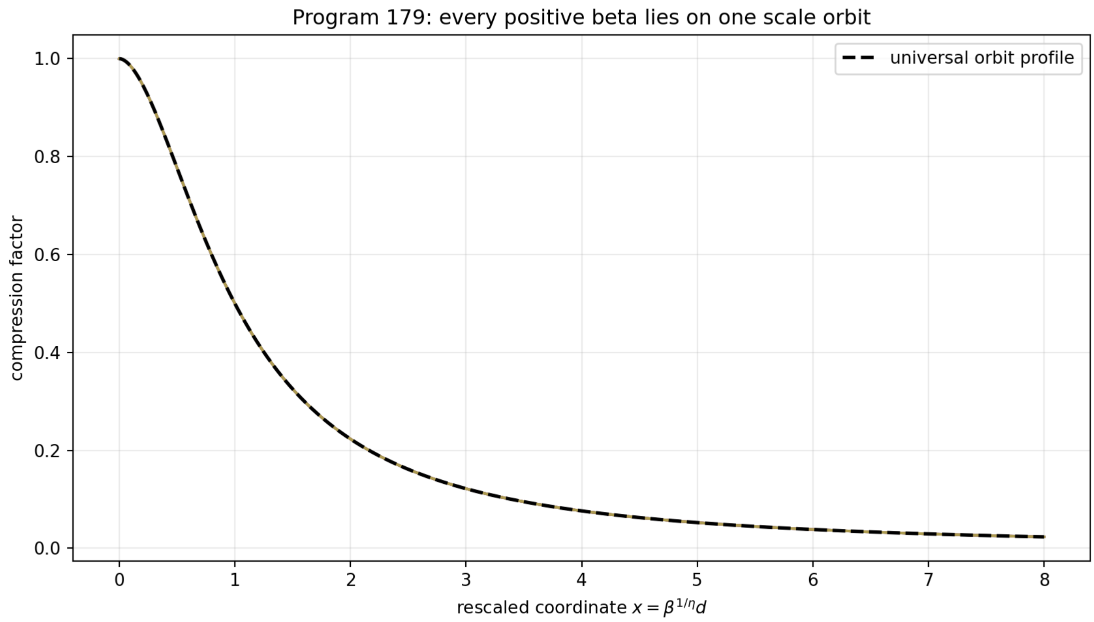

Figure 11: Compression supplies relative scale structure; a calibration torsor remains. An observer inside a layer can measure invariant ratios without access to an absolute external ruler.

The information capacity of a depth-n branch history obeys the elementary lower bound $n \log_2 b$ bits for b freely chosen branches per level. A short self-similar rule can compress a regular history, but it cannot encode an arbitrary universe history at zero cost. FIN's fractal-compression thesis is therefore defensible as *rule-based structural compression*, not as an unrestricted claim that every possible microscopic record fits into a fixed finite string.

14.5 Information independent of a particular medium

The scientifically testable version of "information passes independently of the medium" is representation invariance, not propagation without any realization. Let R_1 and R_2 be two operational realizations and let $\Phi : R_1 \to R_2$ intertwine their preparation, transformation and measurement maps. A quantity $\mathcal{I}$ is medium-independent on this class if

$$\mathcal{I}(R_1, x) = \mathcal{I}(R_2, \Phi x)$$

for every admitted state and intervention. Spectral eigenvalues alone are insufficient: isospectral matrices may have different eigenprojectors, geometry and vertex records. A viable FIN information object must therefore contain at least the relevant spectral projectors, state/preparation family, admissible transformations and measurement context.

The common functional calculus

$$e^{-itA}, \qquad e^{-tA}, \qquad \cos(t\sqrt{A}), \qquad (zI - A)^{-1}$$

shows that one relational structure can be represented in several channels. It does not show that those channels are physically interchangeable. The representation-invariant programme is consequently **[STRONG EVIDENCE]** as a mathematical research direction and **[SPECULATIVE]** as an ontology of nature.

14.6 Near-Haar scale ensembles and the infinite-depth obstruction

A separate study classified scale densities close to log-Haar measure. Trace-class normalization near zero requires

$$\int_0^1 \frac{q(\mu)}{\mu} \, d\mu < \infty.$$

Power softening satisfies this condition for every positive exponent; logarithmic softening does so only beyond its critical exponent. Hence an exactly scale-uniform infinite hierarchy is incompatible with the declared trace-class requirement. One may obtain a broad finite plateau or a normalizable infinite tail, but not both without a trade-off. This is a useful analytic constraint on any future continuum/refinement construction. It supplies neither a Planck cutoff nor a physical layer count.

15 Current FAR state map and the no-replay boundary

The fundamental-action-reconstruction (FAR) programme performed a much broader source audit than the finite ST series. Its current conclusion is not that every route is impossible. It is that the existing artifact set has exhausted several repeated receiver-to-source searches.

15.1 Exact internal structures that survived

The following results are among the strongest current strict or kernel-split-robust exports:

1. The phase cell is exact:

$$\alpha_{\text{geo}} = 4\ln 2, \qquad A_\phi = \frac{2\pi}{\alpha_{\text{geo}}}, \qquad \alpha_{\text{geo}} A_\phi = 2\pi.$$

   Moreover $\alpha_{\text{geo}}/(2\pi)$ is irrational, so $e^{i\alpha_{\text{geo}}}$ is not a root of unity and cannot be a finite $\mathbb{Z}_N$ clock or origin selector.

2. The boundary complex has

$$\operatorname{rank} d_0 = 11, \qquad \dim H^1 = 1, \qquad \omega \overset{\text{inversion}}{\longmapsto} -\omega.$$

   Thus the one-dimensional orientation carrier exists, but its sign is not invariantly selected.

3. A sign-torsor descent system has rank 16 and nullity zero: no simultaneous equivariant $\{\pm 1\}$ section exists in the declared class.

4. Exhaustion of 15,624 admissible weight-zero monomials produced no weight-one positive scale source S_+. Binary origin searches similarly exhausted 4,095 profiles and 351 equivalence classes without an accepted strict source.

5. Only the complete conversion triple $(\ell_*, \tau_*, \hbar_*)$ spans the required rank-three dimension of action density; no strict subset does so.

These are not merely failures to guess a formula. They isolate the mathematical types of missing inputs.

15.2 Current source obligation

The scale-source hitting-set theorem says that every existing candidate route fails the same typed atom: a strict, nonzero, scale-charged source. The least circular admissible schema has the form

$$\text{Source}_{S_+}(\text{nadsoliton}) = s \in V_\chi, \qquad \chi(c) = c, \qquad s > 0,$$

together with an explicit coupling to the dimensional object Ω_M or K_{dim}. No current artifact exports this law. Fitting a dimensional constant, importing a Planck endpoint or reading a scale from a dimensionless invariant would only hide the missing premise.

The corresponding orientation obligation is a non-premise inversion-odd source coupled to the existing chiral receiver. Chiral bispectra, boundary cocycles and memory-lag commutators can detect a supplied sign; they are receivers, not its source. This distinction is the reason QW-2191 remains open.

15.3 Legacy* as a useful generator, not a unification theorem

The most recent Legacy* audit gives the family

$$K^*_{\mathrm{legacy}}(d) = A_0 \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}, \qquad A_0, \beta > 0,$$

a strong finite operator-theory score and a low unification score. Its undirected circulant is real symmetric; its positive-part Laplacian gives valid diffusion and unitary functional calculi; its radial profile is recoverable from the circulant class. But:

- it has no unique Dirac, Maxwell, Yang–Mills or Lindblad completion;
- $d_I = -\log |K^*|$ is not a metric, with 216 of 1,320 triangle tests failing at $\beta = 0.01$;
- the finite positive-semidefinite boundary is approximately $\beta_* = 1.0751507762$ on the declared undirected $\mathbb{Z}_{12}$ carrier;
- its period-eight phase is not a $\mathbb{Z}_{12}$ character;
- the amplitude-only fit to strict has residual at least 0.605;
- it supplies neither physical units nor continuum, gauge, spin or selector data.

Legacy* is therefore valuable as a family of finite spectral, Markov and signed-information models. It does not replace the canonical intermediate legacy kernel and does not complete the strict bridge.

15.4 The two-package scientific architecture

The current state map supports a clean separation:

$$\boxed{W_0 + \mathrm{CA} + \mathrm{OA} \quad (+\mathrm{SA} \text{ when the observable is oriented})}.$$

Here W_0 is the strict dimensionless informational core; CA is the conversion package; OA is the operational package; SA is a sector/reference package. Their logical roles are

$$\mathrm{CA} \simeq (\ell_*, \tau_*, \hbar_*),$$

$$\mathrm{OA} = (\rho_0, \mathcal{P}, \mathcal{T}, \mathcal{I}, \mathcal{E}, \mathcal{A}_{\mathrm{app}}, \mathcal{R}),$$

for state, preparations, dynamics/clock, instruments, environment, apparatus and record, while SA supplies a declared origin/orientation/chirality and its coupling law.

Calling CA or SA an axiom is honest if it is not claimed to follow from W_0. The resulting theory may still be predictive: an axiom earns scientific content when it is minimal, shared across many observables, and exposed to held-out falsification. It simply ceases to be a strict derivation from the frozen kernel alone.

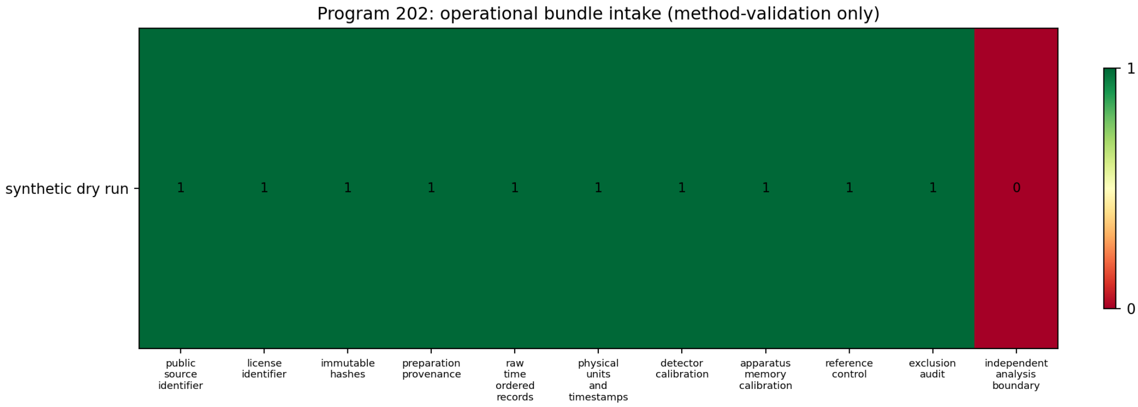

Figure 12: The operational bundle required to turn an abstract generator into a record-producing model. The record/custody component cannot be generated by the same code that proposes the theory.

The current no-new-live-frontier certificate is a stop rule: do not replay generic legacy-to-strict fitting, invariant selector searches, dimensional monomial scans, lower-boundary recursion or reverse-Lagrangian closure without a genuinely new typed object, source law or provider class. Preserving a proved obstruction is more scientific than manufacturing another apparent closure.

16 What the theory now is — and is not

16.1 The deepest surviving mathematical interpretation

Deepest surviving interpretation. FIN is presently a finite, dimensionless spectral-information and contextual-operator framework. A self-adjoint generator and its spectral measure unify strict graph response, Green functions, unitary and diffusive dynamics, variational source response and exact contextual compression. Adaptive learning, orientation, units and physical observation require additional typed laws.

This interpretation is stronger than saying the repository contains unrelated analogies. The Green/spectral/Schur identities are exact. It is also narrower than a field theory, neural universe or Theory of Everything. The current ranking is:

Table 5: Quantitative interpretation ranking, 0–100, based on present theorem coverage.

Interpretation	Score	Confidence	Main limitation
Finite spectral operator / graph signal framework	94	high	finite and dimensionless
Contextual Green–Schur information dynamics	91	high	context split is supplied
Equivariant finite variational theory	88	high	active gain and state space are supplied
Layer-relative observer/refinement theory	85	high	refinement family and instruments are supplied
Operational process/tester research programme	84	high	no admitted laboratory record
Adaptive operator / inference engine	72	moderate	learning/gain law is additional and not uniquely sourced
Information geometry	65	moderate	likelihood/state family and units are supplied inputs
Self-referential dynamical system	60	moderate	feedback integrability and selector remain open
Neural operator / reservoir model	52	moderate	analogy and conditional benchmarks, not ontology
Finite noncommutative geometry	48	moderate	many spectral triples fit the same finite data
Field theory	34	low	no continuum, metric family or complete action
Experimentally established physical theory	8	high	external data and calibrated bridge absent
Theory of Everything	0	high	claim is outside established results

16.2 A complete claim ledger

Table 6: Current FIN knowledge ledger.

Statement	Status	Boundary
Strict $A = sI - W$ is a connected weighted graph Laplacian	**[PROVEN]**	finite frozen $\mathbb{Z}_{12}$ profile
One (A) generates unitary and diffusive calculi	**[PROVEN]**	does not choose physical semantics
Green resolvent unifies spectral response	**[PROVEN]**	$z \notin \sigma(A)$; zero mode handled correctly
Quadratic action reconstructs a supplied Green propagator	**[PROVEN]**	potential/source law not unique
Exact Schur compression preserves retained response	**[PROVEN]**	supplied visible/hidden split
Compression generates strict from legacy	**[REFUTED]**	for tested static binary reductions
Compression generates absolute physical scale	**[REFUTED]**	only relative ratios survive
Shannon data determine energy/temperature	**[REFUTED]**	Gibbs scale nonidentifiability

Statement	Status	Boundary
Chiral receiver detects supplied orientation	**[PROVEN]**	selector sign remains unsourced
Legacy* corrected class follows from audited historical roles	**[PROVEN]**	class-level; A, β not uniquely sourced
Complete legacy-to-strict bridge	open/no-go on current artifacts	phase, nonlinear damping, amplitude and selector sources missing
Adaptive fixed-target flow has Lyapunov function	**[PROVEN]**	fixed covariance and declared projection
Strict-source self-learning law	**[SPECULATIVE]**	feedback integrability/provenance absent
Passive spectral feedback sources active gain g	**[REFUTED]**	wrong sign in the tested passive/Stieltjes classes
Uniform state is uniquely global for $0 \leq g \leq 2.8934$	**[PROVEN]**	supplied functional V_g; adaptive-cover certificate
First global transition lies in $[2.8934, 2.9024964917477667]$	**[PROVEN]**	attaining orbit and uniqueness inside the strip remain open
Exactly twelve global minima at supplied $g = 4$	**[PROVEN]**	one D_{12} orbit; degeneracy is not a selector
Coarse observer has a 78-dimensional blind fiber	**[PROVEN]**	declared refinement and coarse instrument class
Internal clocks determine relative modal ratios	**[PROVEN]**	absolute seconds and cross-channel unit maps remain supplied
Calibration data select a canonical $(\ell_*, \tau_*, \hbar_*)$	**[REFUTED]**	free $(\mathbb{R}_+)^3$ torsor; no invariant section
Degree-six invariant relation kernel has dimension 16	**[PROVEN]**	declared D_{12} invariant basis; quotient dimension 55
Twelve-state cyclic carrier	**[PROVEN]**	"physical octaves" interpretation not derived
Thirty physical fractal layers / Planck bridge	**[REFUTED]** as established	historical external anchor
Particles, masses, gauge fields and gravity from the kernel	**[REFUTED]** as established	required structures absent
Operational experiment specification	**[PROVEN]**	design/software scope only
Laboratory confirmation	absent	independent P241 bundle missing
Exact reduced three-slot comb optimum	**[PROVEN]**	declared q, phase/dephasing model; downstream
P485 causal-axis identity / P487 value relation	open	PREPARE checkpoint is not a result

17 Recommended local research programme

The next programme must not replay closed selector, generic bridge, lower-boundary or Lagrangian lanes without a genuinely new typed object. The following studies are designed for local analytic, exact-algebra and numerical work. They require no external experiment or external audit, although independent verification would remain desirable later.

Table 7: Ranked next local studies. Probability is an estimate of obtaining a useful theorem or decisive no-go, not of confirming FIN physically.

Rank	Programme	Deliverable and falsifier	Prob.	Impact
1	**Strict source law for active gain**	Construct one formula-level, non-circular signed source that exports $g \neq 0$ and couples to the strict state law. Acceptance requires provenance independent of the target minima; a passive reduction proves failure in its class.	0.25	10/10
2	**First-transition orbit and uniqueness**	Resolve the complete strip $[2.8934, 2.9024964917477667]$: identify the first attaining orbit and prove uniqueness or exhibit coexistence.	0.65	9/10
3	**State-sourced selector theorem**	Test whether an internally generated state/orbit can supply an inversion-odd selector without importing an oriented chart. A full D_{12} orbit without a section is a negative verdict.	0.20	10/10
4	**Scale-charged source or explicit CA minimality**	Seek one nonzero weight-one S_+ coupled to $\Omega_M/K_{\mathrm{dim}}$. If none exists in a genuinely new provider class, formalize CA as the irreducible non-strict triple $(\ell_*, \tau_*, \hbar_*)$.	0.35	10/10
5	**Nonlinear nadsoliton existence problem**	Define one nonlinear refinement evolution whose linearization contains strict A; prove or refute a localized persistent coherent state and spectral/orbital stability.	0.30	10/10
6	**Typed dynamic legacy-to-strict completion**	Use a frequency-dependent signed memory map carrying exact legacy phase and strict attenuation. Require transfer to C_{16} and C_{24}; target-only interpolation is rejected.	0.20	10/10
7	**Continuum/refinement-limit functor**	Specify compatible carriers, embeddings, locality norm and convergence topology; prove a nontrivial limit or a compactness obstruction before assigning spacetime or particles.	0.30	10/10
8	**Layer-relative effective-law theorem**	Classify which laws and dimensionless constants are invariant under observer coarse channels, and quantify when enriched instruments recover hidden fibers.	0.65	9/10
9	**Relational spectral clock theorem**	Prove readability, drift bounds and refinement naturality for one recorded mode clock, explicitly separating unitary λ and wave $\sqrt{\lambda}$ scaling.	0.55	9/10
10	**Complete Morse–Conley atlas**	Close the stationary census, indices and connection graph for V_g; any missing boundary orbit falsifies completeness.	0.45	8/10
11	**Exact degree-eight invariant basis**	Construct the forced support-≥ 4 syzygies and compute the exact characteristic-zero quotient; modular rank alone is not a final basis.	0.70	7/10
12	**Adaptive Stieltjes–Schur flow**	Define learning on the self-energy family, test integrability, and prove preservation or necessary violation of passivity. This introduces a new provider class.	0.45	9/10
13	**Synthetic operational identifiability atlas**	Freeze blinded event bundles for unitary, Markov, dephasing and non-Markovian alternatives; quantify required preparations and instruments without target leakage.	0.80	7/10

Rank	Programme	Deliverable and falsifier	Prob.	Impact
14	**Formal core consolidation**	Translate the Dirichlet, Schur, scale, signed-resource and transition certificates into one dependency-minimal formal package. Replay failure exposes hidden assumptions.	0.75	8/10
15	**Bounded P485/P487 resumption**	Only with durable checkpoints, reduce the five exact normal forms and structured eliminant. A nonzero remainder is decisive; PREPARE alone remains no verdict.	0.45	6/10

Programmes 1–7 attack the missing source objects that separate FIN mathematics from physics. Programmes 8–12 deepen structures already shown to be mathematically fertile. Programmes 13–15 improve falsifiability and close older exact-algebra obligations without pretending to create external evidence.

18 Final answer

FIN began with a physically ambitious image: the universe is one self-similar informational soliton that persists through self-coupling, reinforces coherent structure, adapts like a learning network, and gives rise internally to light, matter and observers. That image has not been proved, but it was mathematically productive.

What survives is more precise. The strict finite kernel determines a positive graph generator. Its spectral measure gives wave and diffusion, its resolvent gives response, its Schur complements give exact contextual compression and memory, and its information channels give rigorous distinguishability ledgers. A declared entropy–gain functional now has a certified first global-transition bracket and, at supplied $g = 4$, exactly twelve symmetry-related global minima. Exact observer-blind fibers, relative clocks, invariant relations and infrared continuation make the refinement programme mathematically substantive. The corrected Legacy* formula is the best reconstruction of the old coupling diagram, while the canonical legacy remains an intermediate bridge and strict remains the working completed profile. No theorem yet completes that bridge.

The strongest negative results are equally important. Fractal compression does not generate a physical unit or the strict profile. Shannon information does not determine energy. Passive spectral feedback cannot source the sign of the active gain used in the localization theorem. A twelve-point orbit does not select one of its members. A symmetric radial kernel does not select orientation, and a frozen operator does not learn without an update law. Twelve cyclic states are not automatically twelve physical octaves; thirty historical layers are not a Planck derivation; and the old particle, mass, electroweak and gravity formulas are not current physical consequences.

Therefore the present theory should be described as a *dimensionless spectral-information and conditional variational theory under operational construction.* It contains a coherent mathematical core, rigorous observer-relative reductions and a falsifiable laboratory transfer specification. It does not yet contain the bridge that turns internal phase, relative scale, state and information into unique experimentally calibrated physics. The next breakthrough will not come from another visual analogy or fitted constant. It must export at least one new typed source: an internally justified active/nonlinear law, a non-premise selector, a scale-charged datum, a transferable legacy-to-strict completion, or a continuum/operational map that makes a held-out physical record possible.

A Key equation atlas

The following table is intended as a compact transfer sheet. Each formula is paired with the strongest status licensed by the current record.

Table 8: Principal FIN equations and their epistemic status.

Object	Equation	Status / meaning
Ontology order	nadsoliton $\to$ light $\to$ matter $\to$ emergent observer	Founding programme, not a physical derivation.
Legacy intermediate	$K_{\text{legacy,ont}}(d) = \alpha_{\text{geo}} \cos(\omega d + \phi)/(1 + \beta_{\text{tors}} d)$	Historical/bridge object; no silent role transfer.
Strict working profile	$K_{\text{strict}}(d) = \cos(0.18575d + 0.16250)/(1 + d^{1.8})$	Frozen finite input; phase/damping source remains open.
Corrected Legacy*	$K^*_{\text{legacy}}(d) = A_0 \cos(\pi d/4 + \pi/6)/(1 + \beta d)$	Audited class, not a third final kernel.
Adjacency and row sum	$W_{xx} = 0$, $W_{xy} = K_{\text{strict}}(d(x, y))$, $s = 2\sum_{d=1}^{5} k_d + k_6$	Exact finite construction; $s = 1.660307278766098953\ldots$.
Dirichlet generator	$A = sI - W$, $\langle f, Af \rangle = \frac{1}{2}\sum_{x,y} W_{xy}\lvert f_x - f_y\rvert^2 \geq 0$	Proved; connected graph, $\ker A = \operatorname{span}\{\mathbf{1}\}$.
Dual dynamics	$U_t = e^{-itA}$, $P_t = e^{-tA}$	Proved shared spectral generator; distinct operational meanings.
Wave channel	$C_t = \cos(t\sqrt{A})$	Proved functional calculus; requires displacement/velocity data for a wave equation.
Green response	$G(z) = (zI - A)^{-1} = \int (z - \lambda)^{-1}\, dE_A(\lambda)$	Proved for $z \notin \sigma(A)$.
Source action	$S_\mu[\phi] = \frac{1}{2}\phi^T(A + \mu I)\phi - J^T\phi$	Stationarity gives $(A + \mu I)\phi = J$; inverse reconstruction, not uniqueness.
Operator action	$S[\phi, K] = \frac{1}{2}\phi^T(L + m^2 I)\phi - J^T\phi + \operatorname{Tr} V(K) - \operatorname{Tr}(KC)$	Conditional variational scaffold.
Adaptive law	$\dot{K} = -\Pi(V'(K) - C(K))$	Lyapunov theorem for fixed C; extra derivative term if $C = C(K)$.
Static compression	$\mathcal{C}_H(A) = A_{EE} - A_{EH}A_{HH}^{-1}A_{HE}$	Exact retained zero-frequency response.
Dynamic compression	$A_{\text{eff}}(z) = A_{EE} - A_{EH}(zI + A_{HH})^{-1}A_{HE}$	Exact memory/self-energy object.
Information cell	$\alpha_{\text{geo}} = 4\ln 2 = \ln 16$, $A_\phi = 2\pi/\alpha_{\text{geo}}$	Exact dimensionless identity; not $\hbar$.
Data processing	$D(p\|q) \geq D(Tp\|Tq)$	Proved for stochastic/CPTP contexts in their domains.
Signed path cost	$\mathcal{N}_{\text{path}}^{\min} = 0.406706334692258\ldots$	Certified representation cost; not negative energy.
Variational extension	$V_g(p) = D(p\|u) - \frac{g}{2}(p - u)^T A(p - u)$	Supplied active-gain model; g is not strict-sourced.
First transition	$2.8934 < g_* < 2.9024964917477667$	Certified bracket; first orbit/uniqueness open.
Twelve phases	$\lvert \operatorname{argmin}_{\Delta_{11}} V_4 \rvert = 12$	Proved one D_{12} orbit; not a selector.
Orientation carrier	$\operatorname{rank} d_0 = 11$, $\dim H^1 = 1$, $\omega \mapsto -\omega$	Carrier exists; invariant sign does not.
Conversion package	$\mathrm{CA} \simeq (\ell_*, \tau_*, \hbar_*)$	Minimal rank-three non-strict dimensional input.
Operational package	$\mathrm{OA} = (\rho_0, \mathcal{P}, \mathcal{T}, \mathcal{I}, \mathcal{E}, \mathcal{A}_{\text{app}}, \mathcal{R})$	Necessary state-to-record completion; not generated by A.

B Dependency map: what one object does and does not generate

The exact common-source chain is

$$(A, E_A) \longrightarrow \left\{ \begin{array}{c} e^{-itA},\ e^{-tA},\ \cos(t\sqrt{A}), \\ (zI - A)^{-1},\ f(A),\ \text{quadratic source response} \end{array} \right\}.$$

Once a visible/hidden split is added, this chain extends to Schur response, dynamic self-energy, memory and observer-relative accessibility. Once a state space and the supplied scalar g are added, it extends to the certified equivariant variational transition. None of those arrows exports the following missing objects:

$$\boxed{\begin{array}{c} \text{active source} \oplus \text{selector} \oplus \text{absolute units} \\ \oplus \text{ continuum semantics} \oplus \text{preparation/apparatus/record} \end{array}}$$

This separation is the shortest present explanation of why FIN is mathematically unified yet not physically closed.

C Repository evidence base

This compendium was reconstructed from the local repository, with special reliance on the following evidence classes:

1. `AGENTS.md` and `SUMMARY_GROK.md`: binding scope, provenance and no-go guardrails.
2. `DIAGRAMS_KERNEL_TRANSFORMATION.md`: historical four-mechanism coupling picture.
3. `NATURA_NADSOLITONA.md`: founding ontology and historical physical hypotheses.
4. `K1_LEGACY_ONTOLOGICAL_KERNEL_VS_STRICT_GATE_KERNEL_SPLIT_NOTE.md` and `K2_STRICT_GATE_KERNEL_DERIVATION_CHAIN_NOTE.md`: kernel split and strict provenance.
5. `F2_STRICT_GATE_KERNEL_PROVENANCE_AND_FAR_INPUT_CLASSIFICATION_PACKET.md`, `F3_CURRENT_FAR_FRONTIER_KERNEL_ARTIFACT_SENSITIVITY_CLASSIFICATION_PACKET.md`, and `S2_CURRENT_FAR_STRATEGIC_PRIORITY_REORIENTATION_PACKET.md`.
6. `FIN_Programs_41_50_Legacy_Star_Audit_and_Reconstruction.md`: corrected Legacy* class and negative-coupling audit.
7. `FIN_Programs_51_60_Green_Fractal_Physics_Monograph.md`: Green response, exact compression and dimensional no-go.
8. `FIN_Ten_Research_Programs_Monograph.md`: minimal core, adaptive law, thermodynamic obstruction and operational studies.
9. `FIN_Second_Generation_Atlas_Derived_Nadsoliton_Objects_EN.md`: Stieltjes–Schur context functor and derived-object atlas.
10. `FIN_Dual_Dynamics_Operational_Completion.pdf`: strict Laplacian, exact spectrum, dual dynamics and finite double slit.
11. `FIN_Laboratory_Transfer_Package_P240_242.pdf`: laboratory design, validator and one-shot pipeline.
12. Releases 10.27–10.47: signed resources, operational completion, exact causal-tester mathematics and stopped algebra campaign.
13. `FIN_Singular_Checkpoint_after_PREPARE.zip`: inspected exact-rational remote-runtime preparation artifact.
14. Releases 10.63–10.87 and their JSON/CSV/figure packets: symmetry breaking, refinement, observer geometry, certified variational transitions, invariant algebra and infrared continuation through ST461.
15. The FAR `P3170`, `P3171`, and `P3172` packets: current scale-source obligation, dual-dynamics, and Legacy* operator audits.

D Reproducibility and nonclaims

This release adds no experimental data and makes no network-dependent literature claim. The inspected tracked baseline is git commit `bb540a76`; the compendium also audits named local, not-yet-tracked release packets through ST461 and the current generated FAR records. It does not imply that the worktree was clean. It does not alter the kernels, discharge QW-2191, source active gain or an absolute unit, prove a physical clock, complete legacy-to-strict, transfer legacy physical roles, derive L_{total}, the Standard Model, general relativity, quantum gravity or a Theory of Everything. It does not promote the Singular PREPARE archive to a Gröbner-basis result.

All figures are pre-existing repository artifacts generated by earlier declared scripts. The PDF source is this LaTeX file and is compiled with pdfLaTeX.